

Project Submission Details

Project Title: LiveDoc: Real-Time Intelligent Document Digitizer

Group Details

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1. Description

LiveDoc is an innovative Real-Time Intelligent Document Digitizer built leveraging Computer Vision, Optical Character Recognition (OCR), and Document AI. Moving beyond conventional document scanning—where physical documents are merely captured as flat images or PDFs—LiveDoc aims to intelligently understand the document's internal components. The system dynamically observes and locates documents through a live camera feed, assesses quality, and separates the document into meaningful digital components (such as Text, Tables, Diagrams, and Images) while preserving the original structure and information.

2. Objectives

- **Real-Time Processing:** To develop a system that detects, locates, and captures documents in real-time through a live camera feed.
- **Intelligent Quality Assessment:** To provide real-time user guidance for optimal document positioning and automatically select the highest quality frame by correcting perspective distortion.
- **Document Layout Analysis:** To accurately analyze document structures and classify distinct components like text blocks, tables, charts, and images.
- **Data Extraction & Structuring:** To extract textual data using advanced OCR and intelligently organize the digital representation of the document.
- **Data Validation:** To implement robust validation mechanisms that explicitly flag uncertain, inconsistent, or poorly recognized data to ensure high fidelity.

3. Outcomes

- **Enhanced Digitization Pipeline:** A fully functional application capable of digitizing physical documents into structured, editable, and searchable digital formats rather than static images.
- **Improved User Experience:** An intuitive interface that actively guides users during the scanning process to minimize errors and improve capture quality.
- **Component-Level Extraction:** The ability to not just extract raw text, but also identify and digitally isolate tables, diagrams, and images for targeted downstream use.

- **Reliable Data Conversion:** A highly accurate digitized output with built-in confidence scoring and validation, reducing the need for manual post-scan corrections.